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1.0 UNDERSTAND VARIOUS THEORIES AND MODELS OF THE ATOM

GENERAL INTRODUCTION: Materials are so important in the development of civilization and engineering that we associate ages with them. A proper understanding of materials helps the engineer avoid errors that has so diverted the earlier ages of human existence. Engineering material science seeks to equip the engineer on the properties and structures of materials for efficient, effective and safe design. Understanding of how materials behave like they do and why they differ in properties was only possible with the atomistic understanding allowed by quantum mechanics, that first explained atoms and the solids in the early 1930s. The combination of physics, chemistry and the focus on the relationship between the properties of a material and its microstructure is the domain of material science.

Structure and properties within the context of this study are thus:

Structure at the atomic level is the arrangement of atoms in different ways (gives different properties for graphite than diamond both forms of carbon). Also at the microscopic level the arrangement of small grains of material that can be identified by microscopy (gives different optical properties to transparent Vs frosted glass). While properties are material responds to the environment. For instance, the Mechanical, Electrical and Magnetic properties are responses to Mechanical, Electrical and Magnetic forces respectively. Other important properties are thermal (transmission of heat, heat capacity), optical (absorption, emission and scattering of light) and chemical stability in contact with the environment (like corrosion resistant) eg. Optical communications require fibers that absorb light negligibly. Use of nuclear energy requires solving problems of the waste.

In summary, engineering material science prepares the engineer in a world of variety of countless materials:

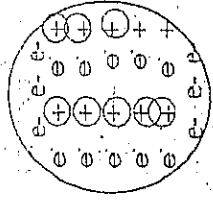
1. To be able to select a material for a given application based on considerations of cost and performance.
2. Be able to create a new material that will have some desirable properties.
3. Understand the limits of materials and the change of their properties within a specific application.

These summaries above now answers world class questions such as:

- Will the materials in use withstand mechanical, electrical and thermal forces and if yes to what extent
- Is the material malleable, and to what extent.
- Can the efficiency of the material increase at high temperatures.
- Will the properties of the material alter appreciably or negligibly during service.
- Will the material be adversely affected by the environmental conditions and resist corrosion and often forms of attack.
- Will the material be acceptable on aesthetic ground? Industrial, Ergonomy and Man-machine relationship/interactions.
- Is the material reliable and to what extent.
- Is the material durable
- Can we produce within acceptable cost while matching safety and other environmental legislations.

1.1 DESCRIBE THE RUTHERFORD - THOMSON ATOM MODELS

In the early 1900s the elements of the atom were already discovered to consist of subatomic particles called Protons and Electrons. However, it was not clear how these protons and electrons were arranged within the atom. J.J Thomson suggested the "plum pudding" model. In this model the electrons and protons are uniformly mixed throughout the atom.

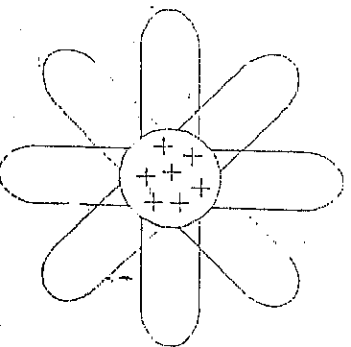


J.J. Thomson's Plum Pudding Models of the Atom.

In this model, electrons and protons are assumed to be in a state of equilibrium and even when excited move or vibrate around their equilibrium positions.

Rutherford, a former student of Thomson in 1909 in a strange experience when every imagined the atom as a "plum pudding", suggested the "planetary model" consisting of a positively charged relatively massive nucleus surrounded by electrons in circularly motion. This finding came as part of an experiment with X-rays, he was shooting a beam of Alpha particles (or alpha rays, emitted by the radioactive element radium) at a sheet of gold foil only 1/3000 of an inch thick and tracing the particles paths. Most of the particles went right through the foil which would be expected if the atoms in the gold were like the plum pudding. But every now and then, a particle bounced back as though it had hit something solid. After tracing many particles and saw that majority of the alpha particles were deflected through small angles while very small number (about 1 in 8000) were deflected through angles larger than 90° and up to 180°. Rutherford reasoned that the way

the alpha particles could be deflected backwards was if most of the mass and positive charge in an atom was concentrated in a nucleus about 10,000 times smaller than the atom itself. All of the negative charges was held in electrons which must orbit the dense nucleus like planets around the sun.



1.2 EXPLAIN BOHR'S THEORY OF THE HYDROGEN ATOM

Niels Bohr in 1913 got an approval from Rutherford to use his laboratory to work on his planetary model since there were some short comings with his earlier discovery. These short comings included it's short lifetime, and discrete spectra emitted by excited atoms. The problems as stated above with the Rutherford atomic model is that, if the electrons are in circulatory motion around the nucleus, then according to classical Physics, it should be ~~very~~ unstable. For one thing, the orbiting electrons should give off energy and eventually spiral down into the nucleus, making the atom collapse. On the other hand, the electrons could be knocked out of position if excited or if a charged particle passed by. Since this never happens Bohr used quantum theory to explain the stability of most atoms. He found that the ratio of energy in electrons and the frequency of their orbits around the their orbits around the nucleus was equal to Planck's constant. Bohr then suggested the revolutionary idea that electrons "jump" between energy levels (orbits) in a quantum fashion; that is without ever existing in an in-between state. Thus when an atom absorbs or gives off energy (either as light or heat), the electron

jumps to higher or lower orbits. According to Bohr, when an atom is excited by the input of energy an electron moves to an orbit of greater radius, subsequently, when the electrons moves back to the ground of energy, energy will be given off or released at some specific frequency giving rise to a "spectral line".

E_0 _____
0 _____
0 _____
0 _____
 E_4 _____

 E_3 _____
 E_2 _____
 E_1 _____

Bohr summarized his postulates in four principles namely:-

- (1) Electrons assume only certain orbits around the nucleus. These orbits are stable and are called "Stationary Orbit".
- (2) Each orbit has an energy associated with it. For example the orbit closest to the nucleus has an energy E_1 the next E_2 and so on.
- (3) Light is emitted when an electron jumps from a higher orbit to a lower orbit to absorb when it jumps from a lower to higher orbit.
- (4) The energy and frequency of light emitted or absorbed is given by the difference between the two orbits Energies.

Eg.:

$$E(Light) = E_{final} - E_{initial}$$
$$= hf$$

Note that only those stationary states for which the orbits angular momentum of the electron is equal to an integral multiple of

$h/2\pi$ are permissible.

ie. $Mvr = nh/2\pi$

Where $n = 1, 2, 3, \dots$ and n is called the principal quantum number and h is Planck's constant. ($h = 6.627 \times 10^{-34} \text{ Js}$).

Unfortunately, Bohr's model worked only for hydrogen. Hence it is most times referred to as Bohr's model of the Hydrogen Atom.

1.3. DERIVE EXPRESSION FOR THE ENERGY LEVELS OF THE HYDROGEN ATOM

Bohr assumed that for the electron to be stable in its orbit, the centrifugal force on it must be equal to the columbic force between it and the electron.

$$\text{Centrifugal force} = mv^2/r$$

Where m = mass, v = Velocity of acceleration, r = radius of the atom or orbital radius.

Hence

$$Ze^2/4\pi\epsilon_0 r^2 = mv^2/r \quad \text{--- (1)}$$

Where Ze is the charge of the nucleus and ϵ_0 is the permittivity of free space.

From (1),

$$Mv^2 = \frac{Ze^2}{4\pi\epsilon_0 r} \quad (2)$$

But kinetic energy $KE = 1/2 Mv^2$ which modifies (2) to be

$$1/2 Mv^2 = -Ze^2/8\pi\epsilon_0 r \quad (\text{ie multiply (2) by } 1/2)$$

The potential energy of the electron is given as

$$PE = -Ze^2/4\pi\epsilon_0 r \quad (PE \text{ is work done in bringing a unit of}$$

charge from infinity to that point)

According to Bohr's model, only certain orbits are permissible and it must be an integral multiple of $h/2\pi$, lets make an attempt to establish the permissible radii of such orbits with reference to the Hydrogen atom.

$$Mvr = nh/2\pi \quad (1)$$

$$Mv^2 = Ze^2/4\pi\epsilon_0 r \quad (2)$$

Squaring equation (1), we have

$$M^2 v^2 r^2 = n^2 h^2 / 4\pi^2 \quad (3)$$

Divide the (3) by m, we have

$$Mv^2 r^2 = n^2 h^2 / 4\pi^2 m \quad (4)$$

Equating (4) and (2), we have

$$\begin{aligned} n^2 h^2 / 4\pi^2 m r^2 &= Ze^2 / 4\pi\epsilon_0 r \\ \Rightarrow n^2 h^2 / 4\pi^2 m r &= Ze^2 / 4\pi\epsilon_0 \quad (\text{multiply both sides by } r) \\ \Rightarrow r &= n^2 h^2 4\pi\epsilon_0 / 4\pi^2 Ze^2 = n^2 h^2 \epsilon_0 / \pi m Ze^2 \end{aligned}$$

Hence the general formular for the permissible radii in the Hydrogen

atom is:

$$r_n = n^2 h^2 \epsilon_0 / \pi m Ze^2$$

substituting the value of r_n into

$$E = -Ze^2/8\pi\epsilon_0 r \quad \text{as obtained above yield a very useful relation for the general Hydrogen atom.}$$

$$\Sigma_n = -Ze^4 M / 8 \Sigma^2 \rho n^2 h^2$$

The energy when the hydrogen atom is stationary is

$$E_1 = -Ze^4 M / 8 \Sigma^2 \rho h$$

This is the energy of the smallest Bohr atom.

Note: The bigger the orbit, the higher the energy.

Also, from

$$E = E_f - E_i \quad \text{we can derive expressions for}$$

transition from one spectral level to the other.

$$\Sigma = hf$$

$$\Rightarrow E_f - E_i = hf$$

$$\Rightarrow F = \frac{E_f - E_i}{h} \quad (1)$$

$$\text{but } E_f = -Ze^4 M / 8 \Sigma^2 \rho n_f^2 h^2 \quad (2)$$

$$\text{and } E_i = -Ze^4 M / 8 \Sigma^2 \rho n_i^2 h^2 \quad (3)$$

equations (2) and (3) are similar, hence can be re-written as

$$E_f = G / n_f^2 \quad \text{and} \quad E_i = G / n_i^2$$

$$\text{Where } G = -Ze^4 M / 8 \Sigma^2 \rho h^2 = 13.6 \text{ eV}$$

Hence

$$E_f - E_i = G / n_f^2 - G / n_i^2$$

$$= G (1/n_f^2 - 1/n_i^2)$$

$$\Rightarrow = \frac{G (1/n_f^2 - 1/n_i^2)}{h}$$

$$= G/h (1/n_f^2 - 1/n_i^2)$$

But $F = c/\lambda$ as stated earlier.

Substituting, we have

$$c/\lambda = G/h (1/n_f^2 - 1/n_i^2)$$

$$1/\lambda = G/hc (1/n_f^2 - 1/n_i^2)$$

$$= -Ze^4 M / 8 \Sigma^2 \rho h^2 hc (1/n_f^2 - 1/n_i^2)$$

$$= RZ^4(1/\pi_f^2 - 1/\pi_i^2)$$

$$= R(1/\pi_f^2 - 1/\pi_i^2) \text{ [but } Z = 1]$$

Where

$R = 1.097 \times 10^7 \text{ m}^{-1}$ is Rydberg constant for a nucleus of infinite mass.

The Bohr model assumes that an electron bound to a nucleus in an atom can only move about the nucleus in certain orbits without radiating energy, but can make a transition from one level to the other, hence Bohr specified the following transition jumps "from one level to the other"

- (1) Lyman Series:- Transition to quantum level $n = 1$ from higher levels.
(ie $n_f = 1, n_i \geq 2$).
- (2) Balmer Series:- Transition to quantum level $n = 2$ from higher levels.
(ie $n_f = 2, n_i > 2$).
- (3) Paschen Series:- Transition to quantum level $n = 3$ from higher levels.
(ie $n_f = 3, n_i > 3$).
- (4) Brackett Series:- Transitions to quantum level $n = 4$ from higher levels.
(ie $n_f = 4, n_i > 4$).
- (5) Pfund Series:- Transitions to quantum level $n = 5$ from higher levels.
(ie $n_f = 5, n_i > 5$) $n_i > \infty$

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These quantum jumps give rise to different radii and spectral lines as shown below.

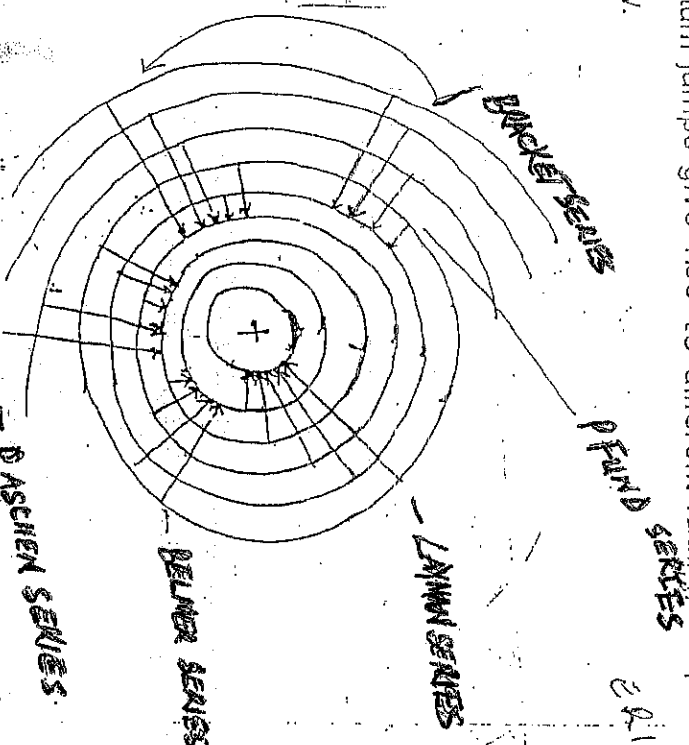


Fig. Circular Bohr orbits for the hydrogen atom showing the transitions which result in the different spectral lines.

1.4 EXPLAIN THE FOLLOWING PHENOMENA IN TERMS OF BOHR'S THEORY

(a) **SPECTRAL LINES**: We have known that there are electrons inside atoms. If the atoms are moved or rapidly shaken or excited, these electrons would give off light and other electromagnetic radiations. Heating separate elements to the extent they glow, would show lines of different wavelengths if the light is passed through a prism. Oh, well we are all with familiar with the rainbow colours of sunlight if viewed through a prism. The rainbow colours is seen because the prism breaks up the light into its separate colours.

Unfortunately, when the light through hydrogen is viewed, the resultant colours weren't the whole rainbow. Instead, bright lines of certain other colours or visible lights. The colours seen is a function of the wavelength and frequency the atom emits its way. These coloured

lines are what we refer to as "SPECTRAL LINES". Just like we have separate signature, so does every atom have its own signature. Scientist can actually tell what an element is by reading the spectral lines. This is a special branch of science known as SPECTROSCOPY.

(b) PHOTO-ELECTRIC EMISSION:- The emission of electrons from a metallic substance by electromagnetic radiation incident on the metal surface is called Photo-Electric Emission.

(c) ELECTRON TRANSMISSION FROM ONE ENERGY TO ANOTHER:- This is referred to as the transition from one energy level to the other. Photons of energy is absorbed or released in the process.

Example 1: Determine the wavelength of the photon emitted in the process of excitation of an electron falling from an energy 19.6 eV to the ground level of energy -32.8 eV.

Solution:

$$\begin{aligned}
 E &= E_f - E_i \\
 &= E_{\text{ground}} - E_{\text{excited}} \\
 &= (32.8 - 19.6) \text{ eV} \\
 &= (32.8 - 19.6) (1.60219 \times 10^{-19}) \text{ J}
 \end{aligned}$$

$$\text{But } E = hf = hc/\lambda$$

$$\Rightarrow \text{but } c = 3 \times 10^8 \text{ ms}^{-1}$$

$$h = 6.6 \times 10^{-34} \text{ Js}$$

$$\begin{aligned}
 \Rightarrow \lambda &= hc/E = (6.6 \times 10^{-34} \times 3 \times 10^8) / (2.115 \times 10^{-18}) \\
 &= 9.3622 \times 10^{-8} \text{ m}
 \end{aligned}$$

EXAMPLE 2:- Calculate the shortest wavelength in the Pfund series of the hydrogen spectrum given that Rydberg constant $R = 1.09 \times 10^7 \text{ m}^{-1}$. also calculate the longest wavelength.

Solution:-

In the Pfund series, the shortest wavelength occurs when $n_f = 5$ and $n_i > \infty$.

Using the formular

$$1/\lambda = R(1/n_1^2 - 1/n_2^2)$$

$$1/\lambda = R(1/5 - 1/\infty)$$

$$= R(1/5 - 0)$$

$$\text{since } 1/\infty = 0$$

$$= R/5$$

$$\lambda = 5/R = 5/1.097 \times 10^7$$

$$= 4.55788 \times 10^{-7} \text{m}$$

(b) The longest wavelength will be when $n_1 = 5$ and $n_2 = 6$

Hence,

$$1/\lambda = R(1/5 - 1/6)$$

$$= R(1/30)$$

$$\Rightarrow \lambda = 30/R = 30/1.097 \times 10^7 \text{m}$$

$$= 2.734 \times 10^{-6} \text{m}$$

Note: $E = 1/\lambda$, hence the Bigger the atom, the smaller the wavelength of light or heat emitted.

CHAPTER TWO

2.0 UNDERSTAND EINSTEIN AND PLANCK'S PHOTO-ELECTRIC

EQUATIONS AND THEIR APPLICATIONS IN ELECTRICAL

ENGINEERING PROBLEMS.

2.1 STATE AND EXPLAIN EINSTEIN'S RELATION

Einstein in his relativistic studies propounded two relations which are

$$E_0 = mc^2 \quad \text{Energy of a particle at rest}$$

$$E = hf \quad \text{Energy of a massless particle}$$

Where m = mass of particle

f = frequency of particle

Example: Find the energy of an electron at rest.

where $m = 0.910 \times 10^{-30} \text{ kg}$

$$\begin{aligned} \text{Solution: } E_0 &= mc^2 \\ &= 0.910 \times 10^{-30} \times (3.0 \times 10^8)^2 \\ &= 0.910 \times 10^{-30} \times 9.0 \times 10^{16} \\ &= 8.19 \times 10^{-14} \text{ J} \\ &= \frac{8.19 \times 10^{-14}}{1.6 \times 10^{-19}} \text{ eV} \\ &= 0.512 \text{ MeV} \end{aligned}$$

Determine its rest

Example 2: A proton has a mass of $1.67 \times 10^{-27} \text{ kg}$. Determine its rest

energy in (i) Joules (ii) MeV. If the proton is moving with velocity $0.5c$,

find (a) its kinetic energy (b) its total energy in MeV.

$$\begin{aligned} \text{Solution: (i) Rest energy, } E_0 &= mc^2 \\ &= (1.67 \times 10^{-27} \text{ kg}) (3 \times 10^8 \text{ ms}^{-1})^2 \\ &= 1.503 \times 10^{-10} \text{ J} \end{aligned}$$

(ii) Rest energy in MeV

$$\begin{aligned} &= \frac{1.503 \times 10^{-10}}{1.6 \times 10^{-19}} \times 10^6 \text{ MeV} \\ &= 939.375 \text{ MeV} \end{aligned}$$

(a) Kinetic energy, KE = $\frac{1}{2} mv^2$

$$= \frac{1}{2} (1.67 \times 10^{-27}) (0.5c)^2$$

$$= \frac{1}{2} (1.67 \times 10^{-27}) (0.25) (3 \times 10^8)^2$$

$$= 1.87875 \times 10^{-11}$$

$$\frac{1.87875 \times 10^{-11}}{1.6 \times 10^{-19} \times 10^6} = 1.17422 \times 10^3$$

$$= 1174.22 \text{ MeV}$$

(b) Total energy = $E_0 + KE$

$$= (939.375 + 1174.22) \text{ MeV}$$

$$= 2113.6 \text{ MeV}$$

2.2 COMPARE EINSTEIN RELATION WITH PLANCK'S EQUATION

Max Planck's developed the blackbody emission and proposed that radiation at certain frequency, F , could be excited only in discrete steps of energy of magnitude $E = hf$.

Albert Einstein confirmed Planck's constant was a fundamental constant of nature. Einstein further gave an explanation to photoelectric effect which classical theory never explained. He modified Planck's equation to be $E = hf - \phi$ where ϕ is the work function of the material. Einstein extended Planck's idea and proposed that all electromagnetic radiation comes in photons of energy. For this reason $E = hf$ is generally known as Einstein's Relation. (ie the photo-energy equation).

NOTE: Work function is the difference in energy between the Fermi level of a solid and the energy of free space outside the solid. (ie the vacuum level). It is the minimum energy required to liberate an electron from the solid at absolute zero temperature. Work function is measured in electron-volt. eV .

2.3 EXPLAIN THE PHOTOELECTRIC EFFECT

Albert Einstein assumed that light was transmitted as a stream of particles termed photons of light at a certain frequency and cause the ejection of electrons from a metal. When a photon strikes the surface of a metal, it raises the energy level of the metal and causes the ejection of electrons. The energy of each photon is given by hf , provided that hf exceeds the work function of the material, ϕ , an electron in the material absorbing a photon is ejected from the surface.

$$E = \frac{1}{2}mv^2 = hf - \phi$$

Regardless of intensity, light with a frequency below the photoelectric threshold will not cause electrons to be emitted from the metal. Ejection only occurs when the energy overcomes the energy binding the electron to the metal.

NOTE: (1) The intensity of light depends on the number of photons in the light. Accordingly, the number of photons in the light, the more intense or brighter the light.

- (2) At low intensities of light, only a few collisions occur, at higher intensities more collisions and ejection occur.
- (3) Each photon in a low intensity beam of light, however, carries the same energy as photons in a high intensity beam.

This implies that ejection of electrons depends on the frequency of light.

- (4) Prevention of photoelectric emission can be achieved by applying a negative potential to the surface.

- (5) The maximum potential required to prevent photo-electricity

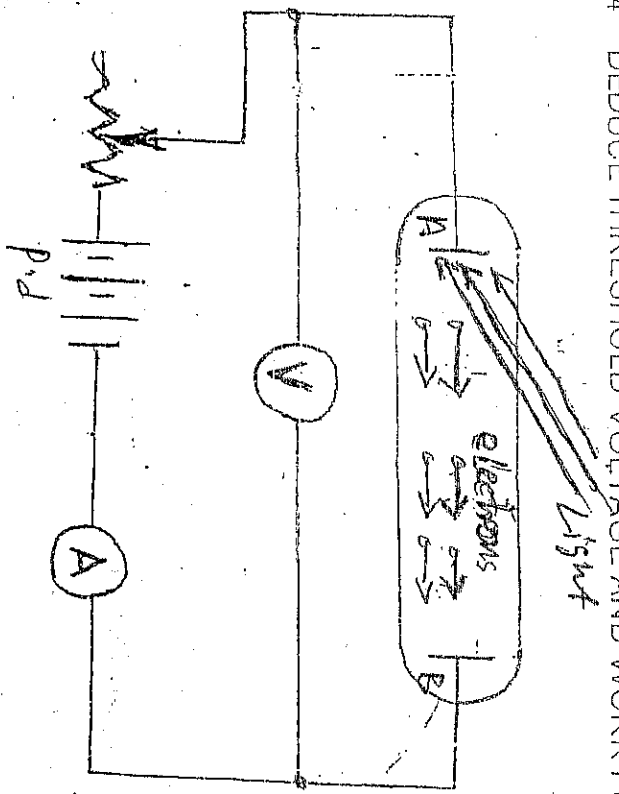
emission is the stopping potential.

Stopping Potential is that voltage that will prevent the photo-electric emission to occur.

Stopping Voltage is that voltage that will prevent the emission of photo-electrons: (V_s)

increased wave frequency $\Rightarrow$ Gr8 m
of signal

2.4 DEDUCE THRESHOLD VOLTAGE AND WORK FUNCTION



The apparatus above was used to investigate photoelectric effect in detail. A voltage V , was maintained between the two electrodes in an evacuated tube. When the electrons leave the surface of the metal, they leave with a kinetic energy.

$$\begin{aligned} \frac{1}{2}mv^2 &= hf - \phi & (1) \\ &= E - \phi & (2) \end{aligned}$$

If E is smaller than ϕ , no electron will be emitted from the surface of the metal. Electrons can only be emitted where $E > \phi$. Not all electrons require the same energy to escape from the surface of a metal.

Hence the maximum value of the kinetic energy of the electron is

$$\frac{1}{2}mv_{\text{max}}^2 = hf - \phi_0 \quad (3)$$

Where ϕ_0 the work function is less than hf .

The maximum kinetic energy can be obtained by applying a potential difference V between the plates as shown. At a particular voltage V_0 the current as read by the ammeter suddenly drops to zero (0), meaning that no electron can be emitted (even the fastest one) from the surface of the metal. This voltage at which this occurs is referred to as stopping voltage or threshold voltage and is given by

$$V_0 = hf - \phi$$

$$V_0 = \frac{hf - \phi}{e} \quad (4)$$

From $eV_0 = hf - \phi$

Note that eV_0 is unit of energy which is the energy acquired by an electron when it passes freely through a potential difference of 1 volt.

Also from Einstein's relation

$$\frac{1}{2}mv^2 = hf - \phi_0$$

If $v = 0$, then

$$\phi_0 = hf \quad (5)$$

i.e. if velocity is = zero

Einstein used the photoelectric effect to disprove some classical theories

which includes

(a) Light is only a wave against the fact in quantum theory ^{that} light is

both a wave and a particle.

(b) The kinetic energy of the emitted electrons should increase with intensity of incident light against the quantum mechanics that the energy is independent of the intensity of incident light, but that it is only related to the frequency of the incident light.

(c) Photoelectric effect should occur at any frequency of the incident light provided that the light is intense against the fact the photoelectric current or number of electrons emitted per second is proportional to the intensity at a particular frequency. For this metal, this frequency is called threshold frequency.

(d) There is time lag for an electron to absorb energy from a very feeble source of light against the fact that there is no time lag between radiation of the surface and the ejection of electrons.

With the help of a diagram explain what you mean by threshold or stopping voltage.

What is work function $\Rightarrow$ the minimum energy required to emit an electron from its shell.

What is threshold frequency.

25. APPLICATION OF PHOTOELECTRIC EFFECT

- (1) Photocells are ~~used~~^{used} in many industrial applications including industrial control accounting operations.
- (2) Solar cells utilize the photoconductivity principle.
- (3) Use to reproduce sound in films
- (4) Used in some fire or smoke detectors
- (5) Used in measuring the dept of liquid in an industrial reservoir. (ie the solar systems etc).

$$V = f\lambda$$

$$f = \frac{V}{2\pi r} = \frac{MZe^2 c^4}{4\epsilon_0^2 n^3 h^3} \quad f \propto \frac{1}{n^3 h}$$

When F is high, λ will be low
 $\lambda \propto \frac{1}{F}$

CHAPTER THREE

3.1 EXPLAIN THE UNCERTAINTY PRINCIPLE OF HEISENBERG

The uncertainty principle states that two complementary properties of a system, such as position and momentum, can never be known (measured) exactly at a time (or simultaneously). This postulation helped cement the dual nature of particles (wave-particle duality of light). This means that by Heisenberg uncertainty principle, it is impossible to know (or measure) precisely and simultaneously both the momentum and position of a particle.

Mathematically, the principle is represented as

$$\Delta_x \Delta_p = \frac{h}{2\pi} = \hbar \quad \text{where}$$

Δ_x = uncertainty in position

Δ_p = uncertainty in momentum

h = Planck's constant

$\hbar$ = Rationalized Planck's constant

The relation shows that the more accurately position is measured (known) the less accurately momentum is known, since their product is a constant.

The 'uncertainty principle of WERNER HEISENBERG becomes a scientific messiah to the problem of atoms' stability.

THEIR APPLICATION TO THE SOLUTION OF PROBLEM IN ENGINEERING

3.2 EXPLAIN THE UNCERTAINTY PRINCIPLES OF HEISENBERG

The uncertainty principle states that two complementary properties of a system, such as position and momentum, can't never be known exactly at a time. This proposition helped cement the dual nature of particles eg. Wave particle duality of light.

This means that by Heisenberg uncertainty principle, it is impossible to know (or measure) precisely and simultaneously both the momentum and position of a particular.

Mathematically, the principle is represented as

$$\Delta x \Delta p = \frac{h}{2\pi} \text{ (where } h/2\pi \text{ is the rationalized Planck's constant, } h.)$$

where Δx is the uncertainty in position and

Δp in uncertainty in momentum.

The relation shows that the more accurately position is measured (known) the less accurately momentum is known. Their product is a constant.

The uncertainty principle of WERNER HEISENBERG becomes a messiah to the problem of atoms stability which classical theory could not explain. Uncertainty principle as applied to microscopic bodies ~~do~~^{do} not find application in macisery objects like a textbook. Light as a particle transfers like a momentum to an object from which it is reflected. Yes oil, because we can see an object in a light which is reflected in own direction. We do not actually feel this transfers momentum, but microscopic objects like the atom receives great disturbance and hence alteration in it position. Another form of representing this principle mathematically is

$$\Delta E \Delta t = \frac{h}{2\pi}$$

Where ΔE is the uncertainty in energy and

Δt is the uncertainty in time

The implication of this uncertainty principle is that a single electron can surround the nucleus of an atom.

Example 1: Find the uncertainty in the position of an electron that has an uncertainty in its speed of 10 ms^{-1} .

Solution: According to the uncertainty principle

$$\Delta_x \Delta_p = \frac{h}{2\pi}$$

$$\text{But } \Delta_p = m\Delta_v$$

$$\therefore \Delta_x = \frac{h}{2\pi\Delta_p} = \frac{h}{2\pi m\Delta_v}$$

$$= \frac{6.6 \times 10^{-34} \text{ Js}}{2\pi \times 9.31 \times 10^{-31} \text{ kg} \times 10^5 \text{ ms}^{-1}}$$

$$= 1.1 \times 10^{-9} \text{ m}$$

Example 2: The position and momentum of an electron is simultaneously determined. If the uncertainty in the position is 10^{-10} m , determine the uncertainty in the momentum.

Solution: From the uncertainty principle

$$\Delta_x \Delta_p = \frac{h}{2\pi} \quad \Delta_x \Delta_p = \frac{h}{2\pi} = \frac{6.6 \times 10^{-34} \text{ Js}}{2\pi \times 10^{-10} \text{ m}}$$

$$= 1.0546 \times 10^{-24} \text{ kgms}^{-1}$$

Example 3: Suppose an electron is the first excited state of hydrogen stays in that state for 10^{-8} sec . Use the uncertainty principle to calculate the width (in eV) of that excited level.

Solution: Using

$$\Delta E \Delta t = \frac{h}{2\pi} \quad \Delta E = \frac{h}{2\pi\Delta t} = \frac{6.6 \times 10^{-34} \text{ Js}}{2\pi \times 10^{-8} \text{ s}}$$

Example 4: If β of an excited speed of electron is 128 ms^{-1} measured to an accuracy of 0.02%, find the uncertainty in position measurement.

Solution:

$$\begin{aligned} \text{Momentum, } p, &= mv \\ &= 9.31 \times 10^{-31} \text{ kg} \times 128 \text{ ms}^{-1} \end{aligned}$$

$$= 1.192 \times 10^{-29} \text{ kgms}^{-1}$$

But momentum was measured to an accuracy of 0.02%, hence

$$\begin{aligned} \Delta p &= 1.192 \times 10^{-29} \text{ kgms}^{-1} \times 0.02\% \\ &= 1.192 \times 10^{-28} \text{ kgms}^{-1} \times 0.02/100 \end{aligned}$$

From uncertainty principle,

$$\begin{aligned} \Delta x \Delta p &= h/2\pi \\ \Rightarrow \Delta x &= \frac{h}{2\pi \Delta p} = \frac{6.6 \times 10^{-34} \text{ Js}}{2\pi \times} \end{aligned}$$

3.3 EXPLAIN PAULI EXCLUSIVE PRINCIPLE

^{3.2} This principle states that no two electrons in an atom can exist in the same quantum energy state, ie, they cannot be described by the same four quantum numbers L, M, N and S.

3.4 APPLY 3.1 TO THE SOLUTIONS OF PROBLEMS ON:

- ^{3.3} (a) Position and momentum of electrons and
(b) Energy and time of electrons transition

^{3.4} 3.5 EXPLAIN THE QUANTUM NUMBERS: To give a complete description of the energy state of electrons, ~~from~~ quantum numbers ~~are~~ depends this

- (a) Principal Quantum Number, n ,:- The principal quantum number, n , gives the main energy level or the distance of the electron from the nucleus. It was positive integer $n = 1, 2, 3, 4, 5, 6$, etc. This numbers gives the exact shell which an electrons is located. These principal shells may be denoted by K, L, M, N, O, P and Q, where $n = 1, 2, 3, 4, 5, 6$ and 7 respectively. The maximum number of electron that can occupy a shell is $2n^2$.

(b) Orbital Angular Quantum Number, L :- The orbital quantum number, l , gives the angular momentum of the electron. Values of l range from 0 to $n-1$1 therefore electrons in the first, second, third and fourth shells can only have angular momentum of 0, (0 and 1), (0, 1 and 2), (0, 1, 2, and 3) respectively.

The values of l are further devoted by letters S, P, D, F where S = 0, P = 1, D = 2, F = 3.

(c) The Magnetic Momentum Quantum Number, m :- The magnetic quantum number, m_l , gives the energy of electrons in an external magnetic field or the magnetic moments of the electron. Notice, electrons produce magnetic fields and have will interact with an external magnetic field. M_l may have any integral value from -1 to +1, including 0. eg. The P shell may have values -1, 0 and +1 and the D shell will have values -2, -1, 0, 1, 2 and 3.

(d) Spin Quantum Number, M_s :- This gives the spin of the individual electrons and has values of $+\frac{1}{2}$ or $-\frac{1}{2}$. Consider the motion within the electron. *They must have different energy state*

3.6 USE PAULI EXCLUSIVE PRINCIPLE AND QUANTUM NUMBER

TO DETERMINE ELECTRON CONFIGURATION FOR ATOMS.

(a) Pauli's principle of exclusion tells us that no two electrons in an atom can possess the same position. Look at an atom in its rest or ground state possesses a question. Simple reasoning would tempt one to say that all electrons possess the same four quantum numbers, thereby bringing to disapprove the Pauli's but a close look shows otherwise. Technically, if they possess the same quantum number, they will invariably be at the

same plane and hence tend to repel each other but ~~not~~ on the converse, the spin values differs. Eg. $n = 1$, or for hydrogen with Atomic number 1.

(0 to $n-1$)

When $n = 1$, ie the K-shell, then $L = 0$ ie s. (0 to $n-1$). Similarly M_L can only be zero (0) (from -1 to L inversely 0).

$M_S = +\frac{1}{2}$ or $-\frac{1}{2}$.

Therefore only two electrons can be accommodated in the K shell. For the $n = 1$, it is in the 1s state ie the first state written as $1s^1$. The one (1) before the s also shows that the electron is located in the K-shell.

Example 2: When $n = 2$ (L-shell) therefore $L = 0, 1$ (orbital quantum number) when $L = 0$, M_L can only be 0.

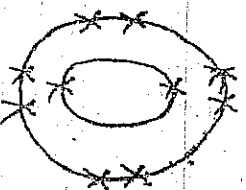
Two electrons in the L shell but when $L = 1$, $M_L = -1, 0$ and $+1$ with two electrons possible in each case, a total number of six electrons can exist in the P state.

ie $M_L = -1$, $M_S = -\frac{1}{2}$ and $+\frac{1}{2}$

$M_L = 0$, $M_S = -\frac{1}{2}$ and $+\frac{1}{2}$

$M_L = +1$, $M_S = -\frac{1}{2}$ and $+\frac{1}{2}$.

Hence, the L shell can contain a total number of eight electrons, two in the 2s state and six in the 2p state.



Atom (Atom)

Example 3: When $n = 3$

(M-shell)

$L = 0, 1$ or 2

(ie d shell)

When $L = 0$, M_L can only be 0 hence $M_S = +\frac{1}{2}$ and $-\frac{1}{2}$

When $L = 1$

$M_L = -1, 0$ or $+1$

Today, I choose to be the best version of myself

With two electron each in this shell finally gives six when $L = 2$.

$$M_l = -2, -1, 0, +1, +2$$

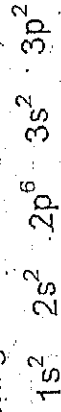
With two electron in each gives 10 electrons.

Hence, the M shell can contain a total number of 18 electrons, two in the 3s state, six in the 3p state and 10 in the 3d state.

Example 4: Write the electrons configuration of an element with atomic number 14 (ie Silicon).

Solution $14 \Rightarrow$ $\begin{array}{cccc} & 2 & 8 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ K & L & M \end{array}$

It's configuration is



Note the general Bohr is

$\begin{array}{l} 1s \\ 2s \ 2p \\ 3s \ 3p \ 3d \\ 4s \ 4p \ 4d \ 4f \\ 5s \ 5p \ 5d \ 5f \\ 6s \ 6p \ 6d \ 6f \\ 7s \end{array}$

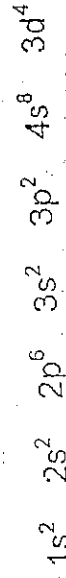
$1s, \ 2s, \ 2p, \ 3s, \ 4s, \ 5s, \ 6s, \ 7s, \ 3d, \ 4d, \ 5d, \ 6d, \ 7d, \ 4f, \ 5f, \ 6f, \ 7f$

Example 5: Write the electronic configuration of Germanium, (Ge) with

atomic number 32.

Solution: $n = 32$

$\Rightarrow$ $\begin{array}{cccc} & K & L & M & N \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 8 & 18 & 4 \end{array}$



(b) CLASSIFY THE ELEMENTS IN ACCORDANCE WITH THE PERIODIC TABLE

The outermost shell electrons, (ie the valence electron) that determines the chemical reactivity of elements only go into reaction to complete for either an octet or duplex outer shell. Elements like Helium, Neon, Argon and Krypton with atomic number 2, 10, 18, 36 and 54 are exactly stable hence chemical inert.

The number of valence electrons determines the group while the number of shells determines the period of any element. Eg. Neon with atomic number 10 has an electrons configuration of $1s^2$ $2s^2$ $2p^6$ and will be expected to be in group zero (0) and in period two. While Silicon would be found in group four (4) and in the third (3^{rd}) period.

Valence electrons are electrons
in the outermost shell.

TABLE VIII
DISTRIBUTION OF ELECTRONS

At	Element	Shells K L M N O	At	Element	Shells K L M N O P
1	Hydrogen	1	No	Cesium	18 8 1
2	Helium	2	55	Barium	18 18 8 2
3	Lithium	2 1	56	Lanthanum	18 18 9 2
4	Beryllium	2 2	57	Cerium	18 18 8 2
5	Boron	2 3	58	Prmium	18 18 8 2
6	Carbon	2 4	59	Neodymium	18 18 8 2
7	Nitrogen	2 5	60	Promethium	18 18 8 2
8	Oxygen	2 6	61	Samarium	18 18 8 2
9	Fluorine	2 7	62	Europium	18 18 8 2
10	Neon	2 8	63	Gadolinium	18 18 8 2
11	Sodium	2 8 1	64	Terbium	18 18 8 2
12	Magnesium	2 8 2	65	Dysprosium	18 18 8 2
13	Aluminum	2 8 3	66	Holmium	18 18 8 2
14	Silicon	2 8 4	67	Erbium	18 18 8 2
15	Phosphorus	2 8 5	68	Thulium	18 18 8 2
16	Sulfur	2 8 6	69	Ytterbium	18 18 8 2
17	Chlorine	2 8 7	70	Lutetium	18 18 8 2
18	Argon	2 8 8	71	Hafnium	18 18 8 2
19	Potassium	2 8 8 1	72	Tantalum	18 18 8 2
20	Calcium	2 8 8 2	73	Tungsten	18 18 8 2
21	Scandium	2 8 9	74	Rhenium	18 18 8 2
22	Titanium	2 8 10	75	Osmium	18 18 8 2
23	Vanadium	2 8 11	76	Iridium	18 18 8 2
24	Chromium	2 8 12	77	Platinum	18 18 8 2
25	Manganese	2 8 13	78	Gold	18 18 8 2
26	Iron	2 8 13 1	79	Mercury	18 18 8 2
27	Cobalt	2 8 13 2	80	Thallium	18 18 8 2
28	Nickel	2 8 13 2 2	81	Lead	18 18 8 2
29	Copper	2 8 13 2 2	82	Bismuth	18 18 8 2
30	Zinc	2 8 13 2 2	83	Polonium	18 18 8 2
31	Gallium	2 8 13 2 2	84	Astatine	18 18 8 2
32	Germanium	2 8 13 2 2	85	Radon	18 18 8 2
33	Arsenic	2 8 13 2 2	86	Francium	18 18 8 2
34	Selenium	2 8 13 2 2	87	Radium	18 18 8 2
35	Bromine	2 8 13 2 2	88	Actinium	18 18 8 2
36	Krypton	2 8 13 2 2	89	Thorium	18 18 8 2
37	Rubidium	2 8 13 2 2	90	Protinium	18 18 8 2
38	Strontium	2 8 13 2 2	91	Uranium	18 18 8 2
39	Yttrium	2 8 13 2 2	92	Neptunium	18 18 8 2
40	Zirconium	2 8 13 2 2	93	Plutonium	18 18 8 2
41	Niobium	2 8 13 2 2	94	Americium	18 18 8 2
42	Molybdenum	2 8 13 2 2	95	Curium	18 18 8 2
43	Technetium	2 8 13 2 2	96	Berkelium	18 18 8 2
44	Ruthenium	2 8 13 2 2	97	Californium	18 18 8 2
45	Rhodium	2 8 13 2 2	98	Einsteinium	18 18 8 2
46	Palladium	2 8 13 2 2	99	Fermium	18 18 8 2
47	Silver	2 8 13 2 2	100	Mendelevium	18 18 8 2
48	Cadmium	2 8 13 2 2	101	Nobelium	18 18 8 2
49	Indium	2 8 13 2 2	102	Lawrencium	18 18 8 2
50	Tin	2 8 13 2 2	103	Rutherfordium	18 18 8 2
51	Antimony	2 8 13 2 2	104	Hahnium	18 18 8 2
52	Tellurium	2 8 13 2 2	105	(Not yet name)	18 18 8 2
53	Iodine	2 8 13 2 2	106	(Not yet name)	18 18 8 2
54	Xenon	2 8 13 2 2	107	(Not yet name)	18 18 8 2
			108	(Not yet name)	18 18 8 2

Note the following point starting from Hydrogen.

- (1) In the first 18 elements the new electrons is always added in the outermost shell until the shell is filled. Then a new shell is started.
- (2) In the higher numbered elements there can be 2 or even 3 unfilled shells.
- (3) Eight electrons temporarily fill each of the shells beyond the M-shell, and a new must be started before more electrons can be filled into the temporarily filled shell.
- (4) There are never more than 8 electrons in the outermost shell.

4.0 UNDERSTAND THE SIGNIFICANCE OF ENERGY BAND AND THEIR

APPLICATIONS

4.1 Explain the energy band in solids.

Individual K, L, M, etc energy levels of an isolated atom are converted into corresponding bands when the atom is in a solid. Though there are a number of energy bands in solids, the following are of particular importance

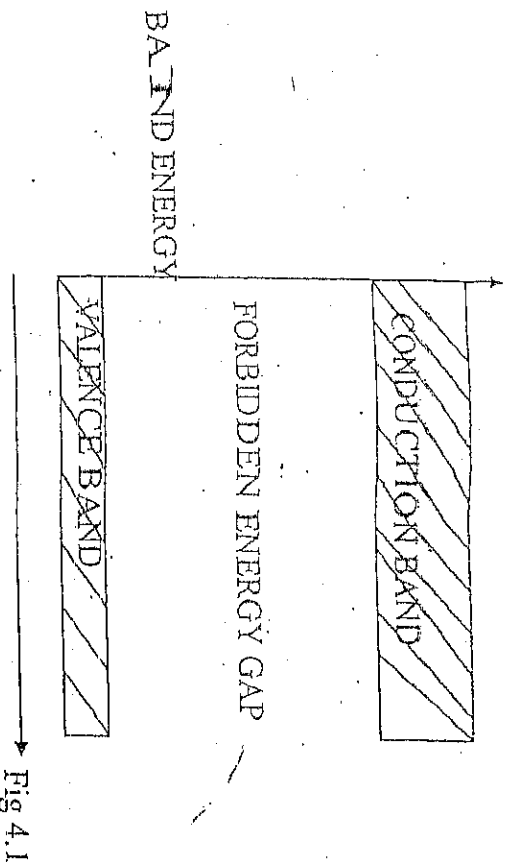
(a) Valence band

(b) Conduction band

(c) Energy gap

(d) Fermi-level.

(a) VALENCE BAND: The range of energies (i.e. band) possessed by valence electrons is known as valence band. The electrons in the outermost or bit of an atom are known as valence electrons. In a normal atom, valence band has the electrons of highest energy. This band may be completely or partially filled. For instance in case of inert gases, the valence band is full whereas for other materials, it is only partially filled. The partially filled band can accommodate more electrons. See below.



(b) **CONDUCTION BAND:** In certain materials (e.g. metals), the valence electrons are loosely attached to the nucleus. Even at ordinary temperature, some of the valence electrons may get detached to become free electrons. In fact, it is ~~use~~ ^{free} free electrons which are responsible for the conduction of current in conductor. For this reason, they are called "**CONDUCTION ELECTRONS**".

The range of energies (ie band) possessed by conduction band electrons is known as **CONDUCTION BAND**.

All electrons in the conduction band are free electrons. If a substance has empty conduction band, it means current conduction is not possible in that substance.

Generally, insulators have empty conduction band. On the other hand, it is partially filled for conductors.

(c) **ENERGY GAP:** The separation between conduction band and valence band on the energy level diagram is known as "**FORBIDDEN ENERGY GAP**". No electron of a solid can stay in a forbidden energy gap as there is no allowed energy state in this

region. The width of the forbidden energy gap is a measure of the bandage of valence electrons to the atom. The greater the energy gap, more tightly the valence electrons are band to the nucleus. In order to push an electron from valence band to the conduction band (ie to make the valence electron free), external energy equal to the forbidden energy gap must be supplied.

(d) **FERMI - LEVEL:** It can be proved that in an intrinsic semiconductor, FERMIE ENERGY LEVEL eg lies in the middle of the energy gap ie indiray between the conduction and valence bands. See diagram below in fig 4.2.

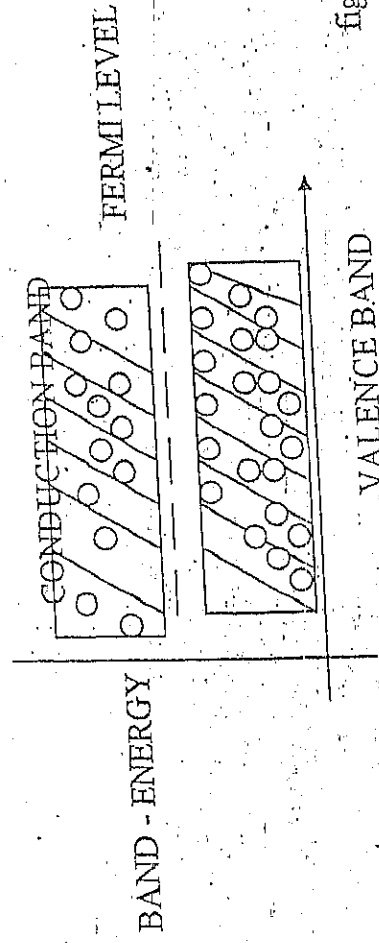


fig.4.2

* (4.2) Explain the properties of conductors, insulators and semi-conductors in terms of energy band structure.

(a) **CONDUCTORS:** Conductors (eg. Copper Aluminum) are those substances which easily allow the passage of electric current through them. It is because there are a large number of free electrons available in a conductor. In terms of energy band, the valence and conduction bands overlap each other as shown in fig 4.3 below. due to this overlapping, a slight potential difference across a conductor ^{causes} the free electrons to constitute electric current. Thus, the electrical behaviour of conductors can be satisfactorily explained by the band energy theory of materials.

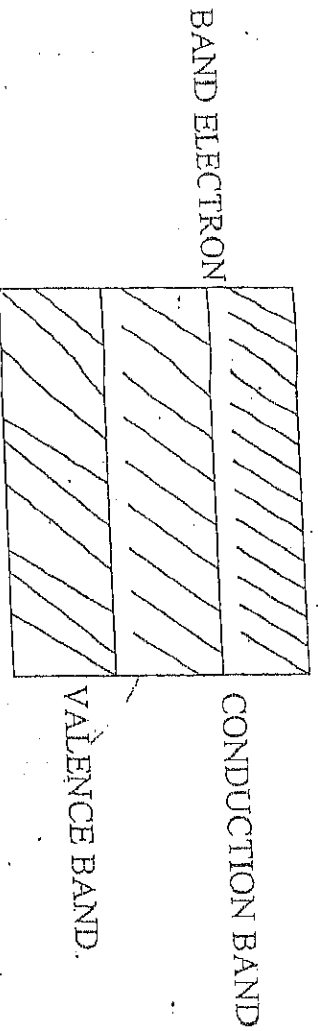


Fig 4.3

Conductor causes the free electrons to constitute electric current. Thus, the electrical behavior of conductors can be satisfactorily explain by the band energy theory of materials

(c) **INSULATORS:** Insulators (eg. Wood, glass etc) are those substances which do not allow the passage of electric current through them. In terms of energy band, the valence band is full while the conduction band is empty. Further, the energy gap between valence and conduction bands is very large (1.sev) as shown in fig 4.4 below. Therefore a very high electric electrons to the conduction band. For this reason, the electrical conductivity of such materials is extremely small and may be regarded as nil under ordinary conditions. At room temperature, the valence electrons of the insulators do not have enough energy to cross over to the conduction band. However, when the temperature is raised, some of the valence electrons may acquire enough energy to cross over to the conduction band. Hence, the resistance of an insulator decreases with the increase in temperature co-efficient of resistance.

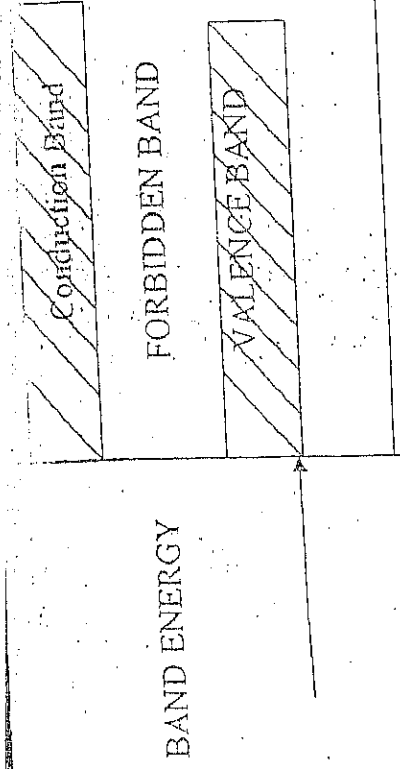


Fig 4.4

(c) **SEMI CONDUCTORS:** Semiconductors (eg. Germanium, silicon etc) are those substances whose electrical conductivity lies in between conductors and insulators. In terms of energy band, the valence band is almost filled and conduction band is almost empty. Further, the energy gap between valence and conduction bands is very small as shown in fig 4.5 below. Therefore, comparatively smaller electric field (smaller than insulators but much greater than conductors) is required to push the electrons from the valence band to the conduction band. In short, a semi conductor has:-

- (a) Almost full valence band.
- (b) Almost empty conduction band
- (c) Small energy gap (lev) between valence and conduction bands. At low temperature, the valence band is completely empty. Therefore, a semi-conductor virtually behaves as an insulator at low temperatures. However, even at room temperature, some electrons (about one electron for 10^{10} atoms) cross over to the conduction band, imparting little conductivity to the semi conductor. As the temperature is increased, more valence electrons cross over to the conduction band and the conductivity increases. This shows that electrical conductivity of a semiconductor increases with the rise in temperature ie a semiconductor has negative temperature coefficient of resistance.

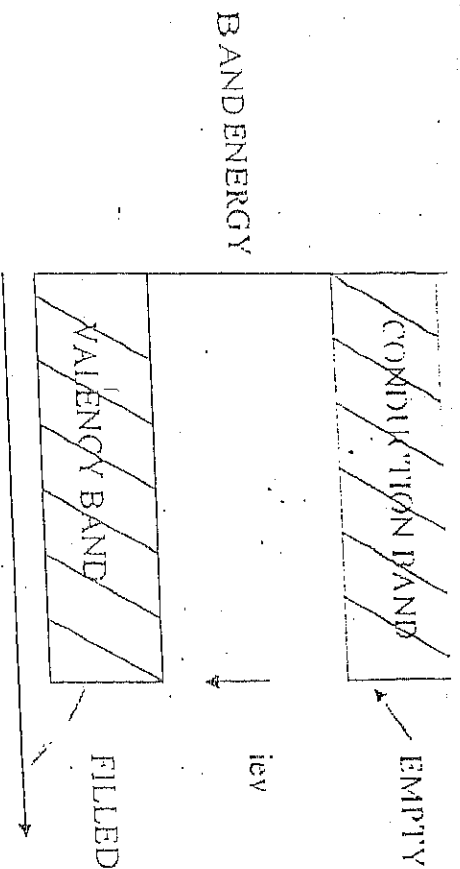
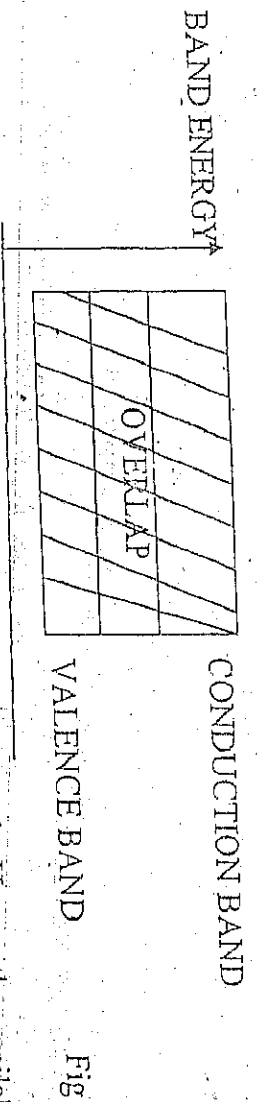


Fig 4.5

4.3 EXPLAINING THE OVERLAPPING OF ENERGY BANDS IN AN ATOM AND THE EFFECTS OF SUCH BANDS.

Put in a simple way, conducting materials are those in which plenty of free electrons are available for electric conduction. In terms of energy bands, it means that electrical conductors are those which have overlapping valence and conduction bands as shown in fig 4.6 below.



Fig

In fact, there is no physical distinction between the two bands. Hence, the availability of a large number of conduction electrons. Another point worth nothing is that in the absence of forbidden energy gap in good conductors there is no structure to establish holes. The total current in such conductors is simply a flow of electron. It is exactly for this reason that the existence of holes was not discovered until semi conductors were studied.

CHAPTER FIVE.

5.0 UNDERSTANDING THE STRUCTURE OF SOLIDS AND THE FORCES WHICH BIND THEM TOGETHER.

5.1 EXPLAIN THE CRYSTALLINE NATURE OF SOLIDS:

The term crystalline, glassy or amorphous will be used to describe solid materials. What do these terms mean? The crystalline state is that in which the

constituent atoms or molecules of the substance are arranged in a regular, repetitive and symmetrical pattern. The regularity of a structure may be termed *LONG*

RANGE, meaning that the same symmetrical pattern of atoms or ions exists over

large distances within the materials, or *SHORT RANGE*, in which symmetrical

groupings of atoms or ions may be in a symmetrical pattern but the relationships between these local groupings relationships between these local groupings may not

be regular. Amorphous, meaning literally without form, is the term used to describe non-crystalline structures, even though these may possess a short range order. Many solids are fully crystalline, some are partially crystalline, and others are amorphous.

Metals and ceramics are crystalline solids, but in many polycrystalline solids, but in

many polycrystalline ceramics there is frequently a glassy phase in the spaces

between crystal grains and a new range of materials, metallic glasses is under

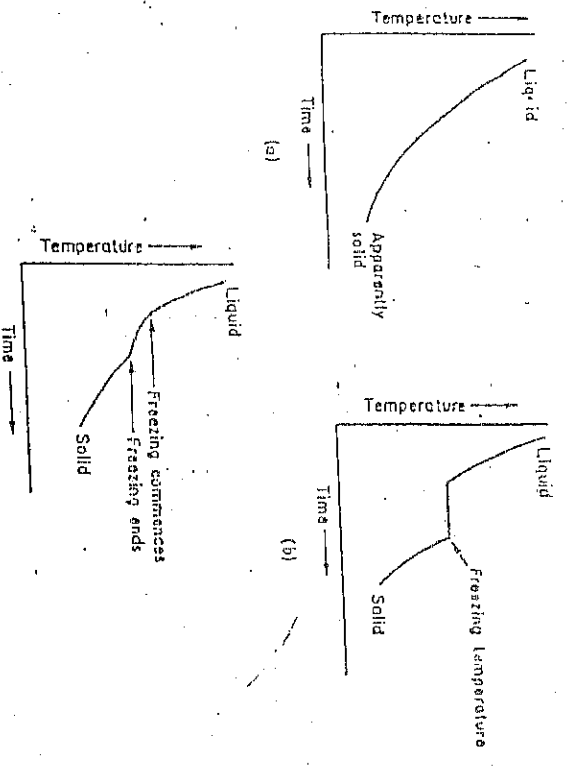
development many polymer materials show a greater or lesser degree of crystalline

but others are amorphous. Inorganic glasses may have the same or similar

compositions as some ceramic materials yet are amorphous rather than being

crystalline.

5.2 CRYSTAL NATURE OF SOLIDS



In solids crystalline state can be defined as that in which the constituent atoms or molecules of the substance are arranged in a regular, repetitive and symmetrical pattern.

If a material is allowed to cool slowly from the liquid state the shape of the cooling curve obtained may be either continuous, or show discontinuities as shown in fig (5.1. a, b, c)

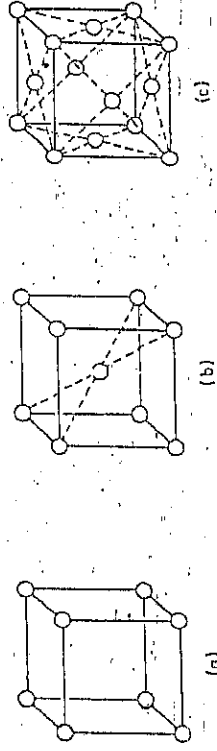
As the temperature falls, there is a steady decrease in the kinetic energy the atoms or

molecules that make up the liquid, and hence a steady increase in the viscosity of the fluid.

Materials with a looking curve as that shown in fig (5.1a) where the atom or molecule still have the same type of random arrangement that existed in the true fluid state. Such materials are amorphous solids or glasses. Other solid materials are crystalline in nature, that is their constituent atoms or molecules are arranged in a definite symmetrical pattern. Fig (5.1b) shows the type of cooling curve obtained for freezing of a pure crystalline substance. This is a very definite freezing or solidification point, shown by marked arrest on the looking curve. At their temperature the atom leave their random movement and tend to stick together in relatively fixed positions in a molecules in a crystalline solid vibrate about fixed positions.

The change from random fluid motion to vibration about a point within a crystal is a change to a much lower energy state. Energy, the latent heat, is emitted from the material at the freezing temperature. Fig(5.1(c)) shows the type of cooling curve applicable to some crystalline mixtures. Freezing is not completed at constant temperature, but takes place over a definite temperature range. The latent heat energy emitted during freezing shows up as a less steep gradient on the cooling curve within the freezing range.

5.3 TYPES OF CRYSTAL STRUCTURES



Unit cells in the cubic system (circles represent atomic centres). (a) Simple cubic. (b) Body-centred cubic (b.c.c.). (c) Face-centred cubic (f.c.c.).

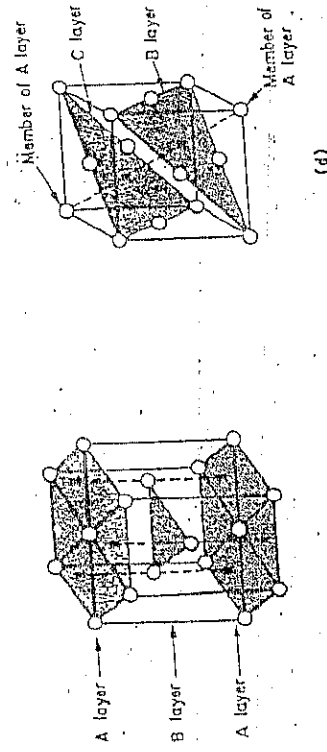
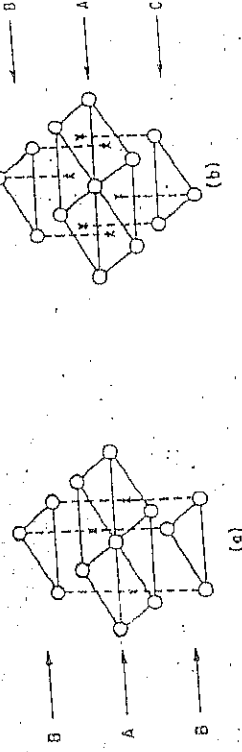
A crystal contains myriads of atoms arranged in a regular, repetitive pattern known as a space lattice. This relative position of atoms within the crystal is shown using unit cell. Unit cell can be regarded as the smallest grouping of atoms still showing the symmetry elements of the type.

This unit cell of the crystal structure can be broadly divided into cubic and hexagonal system. The cubic system can further be divided into cubic, body-centred cubic and face-centred. Atom and ion in the bond state can sometimes be regarded as solid spheres in

contact. In the case of the crystalline form of a pure element, all the spheres will be identical. However when considering a chemical compound or a metallic alloy. Atoms of two or more elements will be present and the different elements will possess differing atomic, or ionic, diameters. In the former case, with identical atoms, the structures tend to have high symmetry. Consequently very many metals crystalline in the cubic or hexagonal form.

Hexagonal close pack is achieved when the atoms are stacked in such a way that the centre of atom in successive layers all lie on a line normal to the base, would give a three-dimensional space lattice in the simple cubic pattern. In this system, each atom is in contact with six other atoms, that is, a co-ordination number of six. The spheres in a simple cubic pattern occupy 52% of the available space.

In the body centered cubic system, the atoms are packed more closely than in the simple cubic form. The closest possible packing of uniform spheres in a plane is hexagonal packing, and the stacking of planes of this type upon one another gives rise to the closest possible packing in three dimensions. It is however, possible to build up a symmetrical three-dimensional structure from close packed planes in two different ways. Fig 5.- shows the centers of atoms (plane A) arranged in a hexagonal plane pattern. When a second plane of atoms is stacked adjacent to the first, or A, plane, the center of atoms in the second layer may be in either the position marked B or that marked C. If three relative spatial positions exist for close packed layers then symmetry in three dimensions can be obtained if the layers are stacked in either an A B A B A B sequence or else in a ABABAB sequence.



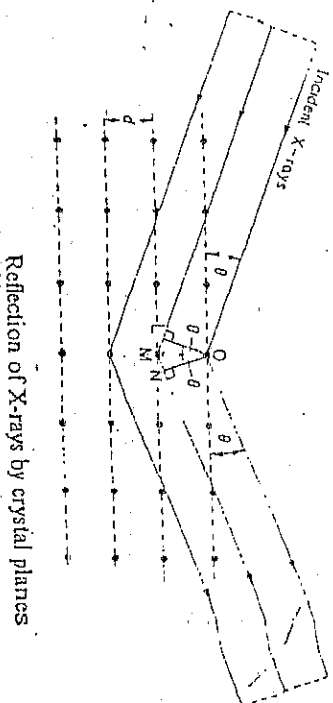
Close packed lattices: (a) ABAB type stacking sequence; (b) ABC type stacking sequence; (c) close packed hexagonal crystal structure: ABAB stacking; (d) face centered cubic crystal system: ABCABC stacking

In both case, the density of packing in the same with 75% of total space filled with spheres and in both case the co-ordination number is 12. the first type of a stacking sequence produces the crystal form known as close packed hexagonal and the second sequence gives rise to the face centered cubic type of crystal.

5.4 X-RAY DIFFRACTION.

The wave length of X radiation are of the same order of magnitude as interlunar spacing in crystals. Consequently, when X radiation is directed at a crystalline material there will be some diffraction ~~effects~~ ^{effects}. The radiation will be reflected by atomic planes, including both surface and sub-surface planes. This is illustrated fig below for radiation striking atomic planes at some angle θ , there will be effective reflection on 1λ when the various ray paths differ by an amount that is equal to one wavelength, or an integral number of wave lengths. Under these conditions the reflected rays will be exactly in phase. From the

diagram this condition will be satisfied when distance LMN is equal to $n\lambda$, when 2 is the wave length of the radiation, and n is a small integer.



$OM = d$, the inter planar spacing

$$\therefore MN = d \sin \theta$$

$$\text{And } LMN = 2d \sin \theta$$

The condition for reflection is therefore $n\lambda = 2d \sin \theta$.

$\Rightarrow$ bragg equation.

By using monochromatic X radiation and measuring the angles at which bragg reflections

occur it is possible to determine the lattice constants for crystals. In the case of cubic crystals of metals the planes that give bragg reflection are:

Body centered cubic - (110), (200), (211), (220), (310), and (222) planes

Face centered cubic - (111), (200), (220), (311), and (222) planes

It will be seen from the above that different types of crystal give different diffraction patterns. For the less symmetrical crystal types, and for those containing more than one type of atom the analysis is similar, but considerable more complex.

e method which is used to obtain X ray diffraction patterns, and hence determine crystal inter planer spacing, in the debye-scherrer method. A powdered metal sample is placed in the centre of a circular diffraction camera and monochromatic X - radiation is directed at it. A strip of film, sensitive to x- radiation, is mounted around the circumference of the camera. The incident X - ray beam is diffracted by crystal planes, according to bragg's law, giving a series of emerging cones of radiation and a series of lines are recorded on the film. The included angle of each radiation cone is equal to $4\theta_n$,

where n is an integer denoting the number of the bragg diffraction. After exposure and development of the film the strip of the film is held flat and the distances between diffraction lines measure. If the distance between a pair of lines is then

$X / R = \theta$ (in radians), where R is the radius of the Camera.

5.6 INTER ATOMIC AND INTER MOLECULAR BONDING.

The fluid and solid substances that we are familiar with are composed of very large aggregates of atom and the properties of these materials derive in part from the manner in which the individual atom are bonded together and the strengths of these bonds. The bonding which crust between atoms is not the same for all materials as there are several types of inter atomic bond possible. Generally, the bonding involves some degree of interaction between the outer shell, or *valence*, electrons and is therefore, dependent on the number and distribution of electrons within the atom. Electron shell confers a very high degree of stability to an atom, this being the electronic structure of the inert or noble gases. In the formation of inter atomic bonds, atoms of elements with a complete outer electron shells attempt, in combination with other atoms, to achieve filled outer electron shells, thus satisfying the condition for stability.

Inter atomic bonds are relatively strong and are term primary bonds. There are other types of bonds, weaker secondary bonds exist between molecules. The principal types of inter atomic bond are the will bond that covalent bond and the metallic bond. Types of inter molecular bond are the hydrogen bond and Vander walls bonds,

5.7 THE IONIC BOND

Element in the lower groups of the periodic table would expose a completely filled electrons shell if they could lose the electrons present in their incomplete outer shell. Group IA elements, the alkali metals, would need to shed only one electron, while Group IIA elements would need to lose two electrons from each atom. Conversely, atoms of the elements in Group VIB and VIIB can complete their unfilled outer electron shell be capturing one or two electrons. An element of Group I, therefore, could react chemically with an element of Group VIIB, resulting in an electron transfer from one atom to the



other, for example, sodium ($Z = 11$) could transfer its one electron from the third principal shell, so exposing a completely filled second shell, to chlorine ($Z = 17$), so giving the chlorine atom the complement of eight electrons in its outer layer (figure below).

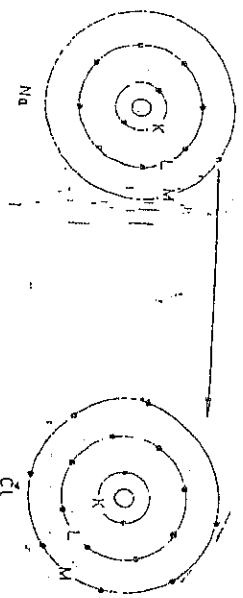
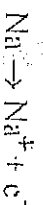
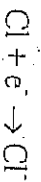


Figure 5.8 Electron transfer from sodium to chlorine

By losing an electron, the sodium atom has become out of balance electricity, and is said to be in ionized state.



Similarly, by gaining an additional electron, the chlorine atom has become ionized



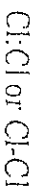
An ion is a charged particle and the net charge may be positive or negative, depending on whether electrons are lost or gained. The magnitude of the net charge may be one, two, three, or in a few cases, even four units, a unit being the charge of a single electron.

5.8 THE COVALENT BOND

In covalent bonding, the stable arrangement of eight electrons in an outer shell is achieved by a process of electron sharing rather than electron transfer. In the case of chlorine, for example, individual atoms combine to form diatomic molecules. The reaction can be written

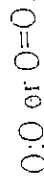


The bond is achieved by the sharing of a pair of electrons. One electron from each atom enters into joint orbit around both nuclei, so giving both nuclei an effective complement of eight outer shell electrons (Figure (a) below). This may be represented symbolically as

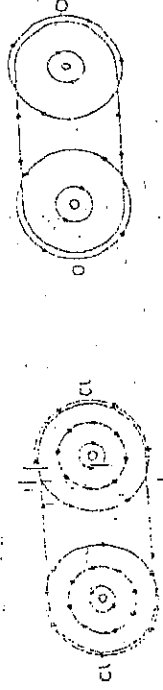


Where a pair of dots, or a hyphen represents a pair of electrons shared between adjacent atoms, namely then, one covalent bond.

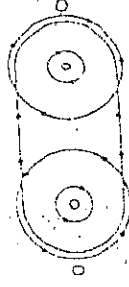
In an oxygen molecule, two pairs of electrons are shared between two adjacent atoms (Figure (b) below) to give each atom a complement of eight outer shell electrons.



In a similar manner to that described for the ionic bond, covalently bonded atoms are subject to both attractive and repulsive forces and the energy of the bond can be represented by the Mie equation with $m = 12$.



(a) Chlorine molecule Cl_2 . One pair of electrons shared.



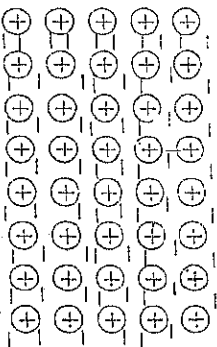
(b) Oxygen molecule O_2 . Two pairs of electrons shared.

Covalent bonds in (a) chlorine and (b) oxygen

5.9 The Metallic bond.

In the bond types discussed so far, the electron distribution is rearranged to provide each nucleus with an external shell containing eight electrons. In the case of metals, where the number of valency electrons to each atom is small, it is not possible to fully satisfy the requirement of eight outer shell electrons per atom, and there is only a partial satisfying of this condition. In an assembly of metal atoms, the principal forces acting will be an attraction due to gravity and a repulsion due to negatively charged electron shells in close proximity. For two atoms, the variation in energy with separation distance, for minimum energy can be regarded as an atomic diameter, and for the discussion of many properties of metals, it is feasible to consider metal atoms as hard spheres of finite radius. At the equilibrium separation, the outer, or valence electron shells of the atoms can be regarded as in contact or slightly overlapping. The outer shell electrons, valence electrons follow complex paths around many nuclei. Thus, all valence electrons are shared by all the atoms in the assembly. This has similarities to a covalent bond. But the strength arrangement of

positive ions permeated by an electron cloud or gas (fig. 5.9). the extreme mobility of valence electrons accounts for the high electrical conductivity, and other properties of metals.



The metallic state, positive ions in an electron gas

5.10 secondary Bonds

In addition to the types of primary bond discussed in the preceding sections, there also exist weaker secondary bonds. These are the hydrogen and van der waal's bonds. The bond energies of the secondary bonds, again, are given by the Mie equation.

$$P.E = A/r + B/r^6$$

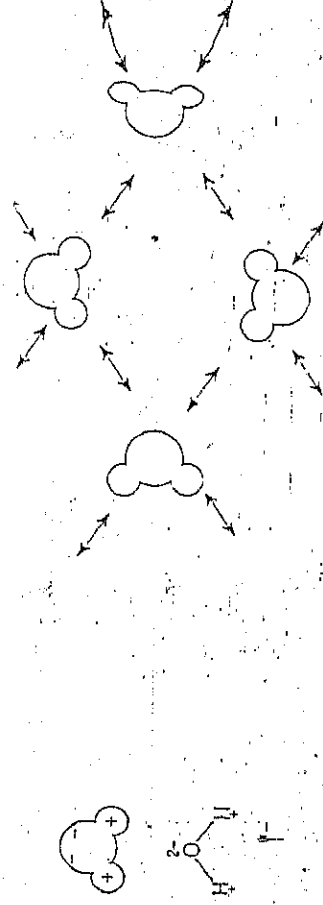
But for these bonds the values of the indices are $m \approx 6$ and $n \approx 12$ and typical bond energies are of the order of 0.2 eV for the hydrogen and between 0.002 and 0.1 eV for van der waal's bonds as compared with 6.5 eV for the ionic bond in sodium chloride. The bonding in the H_2O molecule is polar with the two hydrogen atoms being positive relative to the two non-bonding orbitals of the oxygen Atoms. there is quite a strong force of attraction between the hydrogen and the negative ends of adjacent molecules. The

hydrogen can be written thus: $H-O-H \dots O$

The hydrogen bond does not occur only in water and ice but in a number of polymer materials. In general terms, a hydrogen bond can be formed between an electronegative atom and a hydrogen atom which is already covalently bonded to another electronegative atom. Examples of hydrogen bonding in polymers are $-M-H \dots O$ bonds between polyamide (Nylon) molecules and $-O-H \dots O$ bonds in cellulose and polyvinyl alcohols. Many molecular compounds are polarized to some extent and electrostatic attractive forces exist between the molecular dipoles. These weak electrostatic attractive forces are termed Van der Waal's bonds. Van der Waal's bonds can occur also between atoms. The monatomic inert gases, with full outer electron shells will condense into liquids and solids

at extremely low temperatures. This indicates the existence of weak bonding forces. Within such atoms, owing to the continual movement of electrons at any instant the centroid of negative charge need not coincide with the centre of positive charge, that is, the nucleus (fig. Shown below). the atom becomes slightly polarized and may be weakly attracted to a similar polarized atom.

At very low temperatures, the kinetic energy of the atoms will be insufficient to overcome even these very weak attractions and the gas will condense. The energies of the Vander Waal's bonds in the inert gases is of the bond energy may approach that of the hydrogen bond. The magnitude of the bond energy may approach that of the hydrogen bond.



(a) Representation of a polar H_2O molecule; (b) attraction between neighbouring H_2O molecules—the hydrogen bond

CHAPTER SIX

6.1 Imperfections in Solid and their Effects on Property.

Modern theories of plastic flow in metals are based on the existence of small imperfections, or defects within the crystals. These structural defects, termed dislocations and plastic deformation are due to the movement of dislocations across the slip planes of a crystal under the action of an applied stress.

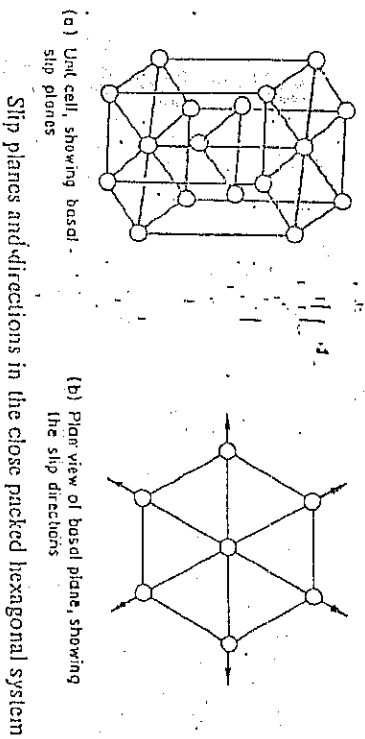
6.2 Slip planes

The crystal planes on which slip generally takes place are those that possess the highest degree of atomic packing. The direction of slip within a crystal plane is the direction of greatest

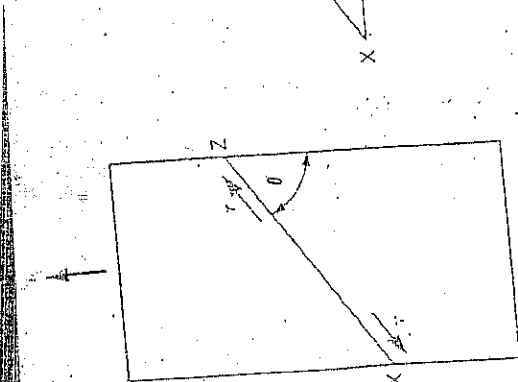
atomic line density. The majority of metals crystallize as close packed hexagonal, face centered cubic, or body centered cubic, and slip within these three systems will be discussed.

In the close packed hexagonal system, the planes with the greatest density of atomic packing are the basal planes (001) and within planes of this type there possible slip directions (Figure below). Within any one hexagonal crystal space lattice there is only one series of parallel planes of this type.

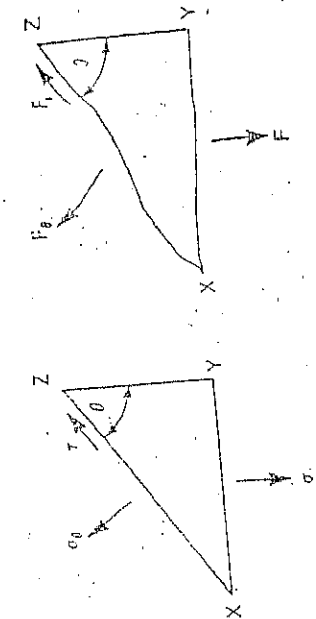
The result of research work in the 1930s into the effects of tensile stresses on single crystal.



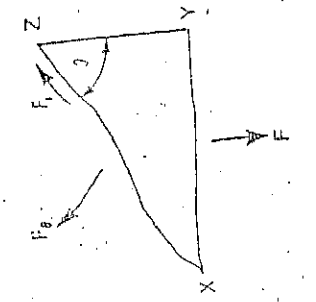
Specimens of zinc, a close packed hexagonal metal, showed that the behavior of the sample was dependent on the angle at which the slip planes were inclined to the axis of the applied stress. If the slip planes were aligned either normal to, or parallel to, the stress axis, failure occurred in a brittle manner with negligible plastic deformation. If however the slip planes were inclined at some angle, ϕ , other than 0° or 90° , plastic yielding took place before failure. The amount of plastic yielding and the value of the direct tensile stress necessary to commence yielding were not constant but varied with angle ϕ .



(a)



(b)

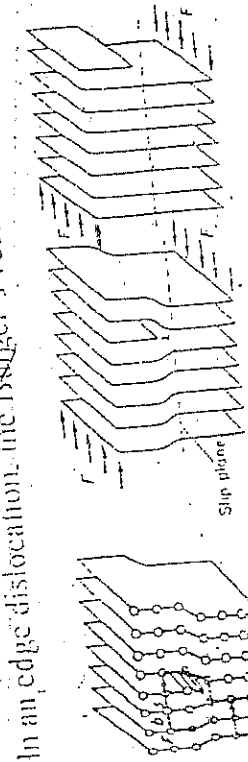


(c)

6.2 DISLOCATIONS

Crystal space lattices are not perfectly crystalline but contain certain structural defects. These may be classified as either or point defects. The line defects are known as *dislocations*, and these may be classified as either *edge-type* or *screw-type*. In practice, dislocation lines are rarely of the pure-edge or pure-screw type but are mixed dislocations, that is, lines of dislocation containing both an edge and a screw component. An important property of a dislocation is its *Burger's vector*, b , which indicates the extent of lattice displacement caused by the dislocation. The Burger's vector also indicates the direction in which slip will occur. Figure 6.3 is a representation of an edge dislocation. It can be considered that the edge dislocation is due to the presence of an additional half-row of atoms within the lattice. If an atom-to-atom circuit is described with in a portion of regular lattice, as shown at the bottom left of Figure 6.3(a), it will be a complete closed circuit. If, however, a similar circuit is described around the dislocated portion of lattice the start and finish will not be coincident. The distance SF in Figure 6.3(a) will be the Burger's vector,

but in an edge dislocation, the Burger's vector is normal to the dislocation line.



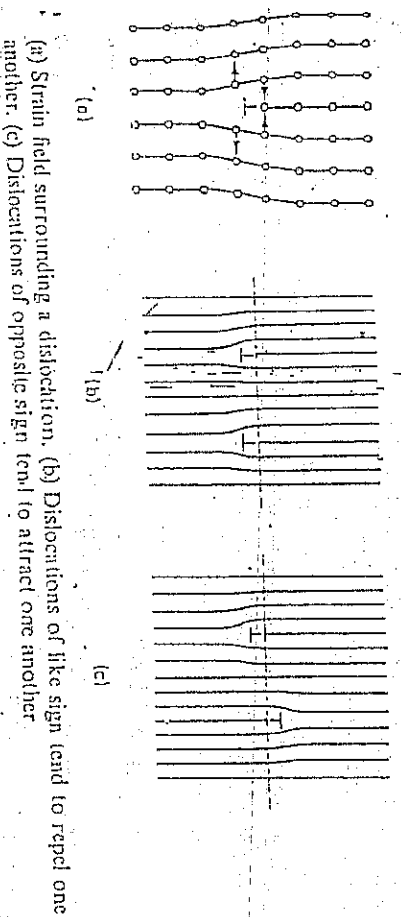
(a)

Figure 6.3(b) shows a 3D representation of a crystal lattice with an edge dislocation. A slip plane is indicated. A circuit is drawn around the dislocation, showing the start and finish points S and F, with the distance SF representing the Burger's vector.

(b)

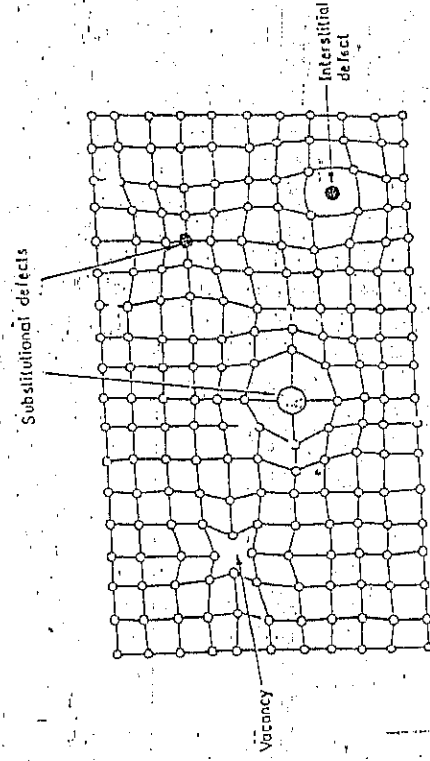
When a shearing stress is applied to a section of crystal lattice, and this stress is beyond the elastic limit, minor atomic movements will occur causing the dislocation line to move through the lattice (~~dislocation~~). The magnitude of the stress necessary to initiate the movement of a dislocation across a slip plane is considerably less than that which would be required to bring about block slip. It will be seen that plastic flow occurs in the direction of the Burger's vector.

It should be appreciated that there will be strain fields associated with dislocations. In the example of an edge - type dislocation, considered as an extra half-row of atoms inserted into the structure, the crystal lattice in the region of the dislocation will be strained. There will be a compressive strain above the dislocation and a tensile strain below it, as shown in fig. (a) bellow. it can be said that dislocations of like sign repel one another. Consider two similar dislocations on the same slip plane (fig(b) bellow. these will tend to repel one another in an effort to reduce the strain in that portion of the lattice, and would require an increase in the level of applied stress to move them closer together. Conversely, dislocations of opposite sign attract one another. If the two dislocations shown in fig (c) bellow were brought together they would cancel one another out leaving a perfect unstrained section of lattice.



(a) Strain field surrounding a dislocation. (b) Dislocations of like sign tend to repel one another. (c) Dislocations of opposite sign tend to attract one another.

Crystals invariably contain numerous point defects in addition to the structural defects mentioned above. The principal types of point defects which occur in metal crystals are the vacancy, the substitutional defect, and the interstitial type defect. The latter two types are due to the presence of 'stranger' atoms of other elements which are present either as impurities or are deliberately added as alloying elements. The interstitial type occurs when the stranger atoms are very small in comparison with the atoms of the parent metal and fit into interstitial sites, while the substitutional type occurs when parent and stranger atoms are comparable in size. It will be noticed from fig. below that strain is developed within a crystal lattice in the neighbourhood of point defects.



Lattice containing several types of point defect. Note that substitutional atoms may be larger or smaller than the parent atoms causing positive or negative lattice strain

CHAPTER SEVEN

III. THERMAL AND OPTICAL PROPERTIES OF MATERIALS

7.1 ABSORPTION AND TRANSMISSION

In metals the outer-shell, or valency, electrons are in partially filled conduction bands. When photons of comparatively low energy are incident upon a metal the photon energy can be transferred to valency electrons, raising the energies of these electrons within the

conduction band. When this occurs the incident energy is absorbed by the material and not transmitted through it. Metals are opaque to radio waves, infra-red, visible light and ultra-violet but the high-energy photons of X- and γ -rays can penetrate through the metals.

There is some absorption of this high-frequency radiation and the total amount of X- or γ -radiation that is transmitted through the metal is inversely proportional to the thickness.

The absorptive power of a metal for X-radiation is also in proportion to the density of the metal the are a metal for X-radiation is also in proportion to the density of the metal. X- and γ -radiation are used for radiography of metals and greater penetration is achieved by the harder (higher-frequency) radiation. There is some reflection of X-rays from atomic nuclei and because of this fact X-rays can be used for investigation of crystal structures.

Polished metal surfaces will largely reflect, rather than absorb incident low-energy photons. This reflection is partially selective. For example, copper absorbs visible light at the short wavelength end of the visible spectrum, hence its reddish colour.

Dielectric materials possess filled electron-energy bands and forbidden energy gaps. There cannot be transference of energy photons to electrons in these materials in the same way as in metals and so these materials generally do not absorb electromagnetic radiation but transmit it. Good radio reception is obtained within a building since the brick and glass are largely transparent to radio waves, but radio reception within a metal-bodied car is difficult without an external aerial.

The explanation of absorption and transmission given is a very simple one and does not cover the situation fully. Non-metallic materials are not fully transparent to all types of electromagnetic radiation and many materials show absorption peaks at particular frequencies. For example, window glass is opaque to the infra-red section of the spectrum.

Also certain materials show definite colours because they strongly absorb certain frequencies within the visible part of the energy spectrum.

These colour effects are caused by the presence of certain types of crystal defects, termed colour centres. Impurity ions in certain materials may create donor or acceptor energy levels which cause selective absorption of photons of particular wavelengths. It is this selective absorption which is responsible for the varied colour of materials. Many types of defect (colour centre) may occur, the best known example being the F-centre which is an electron trapped on an anion vacancy and this has a series of energy levels available to it.

energy required to transfer from one level to another falls within the visible part of the electromagnetic spectrum giving the characteristic colour, which depends on the host crystal. F-centres in sodium chloride, NaCl, give the characteristic greenish-yellow colour. Conductor materials possess filled valence bands but with forbidden energy gaps of a smaller size than those in insulators. Low-frequency radiation is not absorbed by conductors but if the frequency of the incident radiation is such that the photon energy is equal to the magnitude of the energy gap, energy may be transferred from photons to electrons. In other words, the incident energy is absorbed and electrons are excited into the conduction band. This gives rise to the effect termed *photo-conduction* since the conductivity of the material increases in a similar manner to that caused by raising the temperature of the semiconductor.

LUMINESCENCE

Luminescence is the general term covering the phenomenon of light emission from a material as a result of excitation of some type. During excitation, some of the valence electrons may be excited into a higher energy band, as mentioned in section 14.1. When excited electrons drop back to the ground state, or into some intermediate energy level, energy will be emitted from the material. An emission which occurs only for the duration of the excitation is termed *fluorescence*. A delayed emission which persists for a period of time after excitation has ceased is termed *phosphorescence*. The intensity of a phosphorescent emission at some time t after excitation has ceased is I_t , which delays with time according to the expression: $I_t = I_0 \exp(-t/T)$ where I_0 is the initial intensity and T is a constant for the material termed the relaxation time. Phosphorescent materials, or phosphors, are used to coat the surface of television and radar tubes. For a television tube, the relaxation time of the phosphors must not be too long or the images will begin to overlap but for the display tubes of search radar a phosphor with a longer relaxation time is selected to give an afterglow period of several seconds. Excitation of a luminescent material may be caused in one of several ways, by incident photons, termed, *photo-luminescence*; by an incident electron beam, termed *cathodo-luminescence*, by the effects of an alternating electric field, termed *electro-luminescence*, or by certain chemical reactions, termed *chemo-luminescence*.